

Multiple Choice Questions – Network Fundamentals

1. What is the largest computer network in the world?
 - A. LAN
 - B. MAN
 - C. Internet
 - D. PAN
2. What was the predecessor of the Internet?
 - A. Ethernet
 - B. ARPAnet
 - C. WiMAX
 - D. Bluetooth
3. Which wave attribute refers to the number of cycles per second?
 - A. Amplitude
 - B. Frequency
 - C. Period
 - D. Phase
4. What is the SI unit of frequency?
 - A. Volt
 - B. Meter
 - C. Hertz
 - D. Watt
5. Which wave attribute refers to the height above and below the x-axis?
 - A. Frequency
 - B. Wavelength
 - C. Phase
 - D. Amplitude
6. Which type of wave continuously varies in amplitude and time?
 - A. Digital wave
 - B. Analog wave
 - C. Square wave
 - D. Pulse wave
7. Which of the following is an example of a digital signal?
 - A. Human voice
 - B. Music
 - C. Computer memory data (0s and 1s)
 - D. Radio broadcast
8. Which networking media uses cables and is limited by physical geography?
 - A. Wireless media
 - B. Optical media
 - C. Unbounded media
 - D. Wired media
9. Which transmission mode allows communication in only one direction?
 - A. Full-duplex

- B. Half-duplex
 - C. Simplex
 - D. Parallel
10. Which transmission mode allows simultaneous two-way communication?
- A. Simplex
 - B. Half-duplex
 - C. Full-duplex
 - D. Serial
11. Which transmission type sends bits one at a time over a single wire?
- A. Parallel transmission
 - B. Duplex transmission
 - C. Serial transmission
 - D. Analog transmission
12. Which cable uses BNC-T connectors?
- A. Thicknet cable
 - B. Fiber optic cable
 - C. Thin (Thinnet) cable
 - D. STP cable
13. Which coaxial cable type uses vampire taps?
- A. Thinnet
 - B. Thicknet
 - C. UTP
 - D. Fiber optic
14. What does UTP stand for?
- A. Universal Twisted Pair
 - B. Unshielded Twisted-Pair
 - C. Unified Twisted Pair
 - D. Unshared Twisted Pair
15. In twisted-pair cables, what term refers to signal loss due to impedance?
- A. Crosstalk
 - B. Reflection
 - C. Attenuation
 - D. Propagation
16. Which twisted-pair cable includes metallic foil wrapping?
- A. UTP
 - B. Fiber optic
 - C. Coaxial
 - D. STP
17. What is the innermost section of a fiber optic cable called?
- A. Cladding
 - B. Coating
 - C. Jacket
 - D. Core
18. Which fiber optic cable type is best for long-distance high-speed transmission?
- A. Multimode
 - B. Single-mode

- C. Coaxial
 - D. Twisted-pair
19. Which wireless medium is commonly used for radio and TV broadcasting?
- A. Microwave
 - B. Infrared
 - C. Radio wave
 - D. Bluetooth
20. What device converts electrical energy into electromagnetic energy and vice versa?
- A. Router
 - B. Antenna
 - C. Repeater
 - D. Modem
21. In Line-of-Sight propagation, the antennas must:
- A. Be underground
 - B. Use fiber optics
 - C. See each other
 - D. Be connected by cable
22. Which propagation method is used for frequencies from 3 MHz to 30 MHz?
- A. Ground wave
 - B. Space wave
 - C. Skywave
 - D. Infrared wave
23. Which electromagnetic spectrum band is used for commercial AM radio broadcasting?
- A. HF
 - B. VHF
 - C. MF
 - D. UHF
24. Which frequency band is commonly used for FM broadcasting?
- A. ELF
 - B. VHF
 - C. SHF
 - D. EHF
25. Which band is commonly used for satellite communication?
- A. LF
 - B. MF
 - C. SHF
 - D. VLF
26. Which wireless medium is commonly used in remote controls?
- A. Bluetooth
 - B. Microwave
 - C. Infrared
 - D. Radio wave
27. Bluetooth operates mainly at what frequency?
- A. 900 MHz
 - B. 1.2 GHz

- C. 2.4 GHz
 - D. 5 GHz
28. Which transmission method sends multiple bits simultaneously using different channels?
- A. Serial transmission
 - B. Parallel transmission
 - C. Simplex transmission
 - D. Full-duplex transmission
29. Which type of fiber optic cable allows light to travel through numerous paths?
- A. Single-mode
 - B. Multimode
 - C. Coaxial
 - D. Shielded-mode
30. What is the main purpose of a signal in communication systems?
- A. To store files
 - B. To increase cable size
 - C. To maintain synchronization between sender and receiver
 - D. To replace transmission media
31. • Which topology connects all computers to a single central hub?
- A. Bus Topology
 - B. Ring Topology
 - C. Star Topology
 - D. Mesh Topology
32. • In a star topology, what happens if the central node fails?
- A. Only one computer is affected
 - B. The network becomes faster
 - C. The entire network communication is affected
 - D. Nothing happens
33. • Which topology connects all devices to a single cable?
- A. Mesh Topology
 - B. Bus Topology
 - C. Tree Topology
 - D. Hybrid Topology
34. • What is another name for a bus topology with exactly two endpoints?
- A. Circular Bus
 - B. Linear Bus Topology
 - C. Hybrid Bus
 - D. Parallel Bus
35. • Which topology forms a closed loop where each device has exactly two neighbors?
- A. Ring Topology
 - B. Star Topology
 - C. Tree Topology
 - D. Mesh Topology
36. • Which topology is also called a hierarchical topology?
- A. Ring Topology
 - B. Bus Topology

- C. Tree Topology
 - D. Mesh Topology
37. • Which topology provides the highest reliability but is expensive to expand?
- A. Bus Topology
 - B. Full Mesh Topology
 - C. Ring Topology
 - D. Star Topology
38. • In a partial mesh topology, devices are connected to:
- A. Only one device
 - B. Every device in the network
 - C. Some selected devices
 - D. No other devices
39. • Which topology combines two or more different topologies?
- A. Ring Topology
 - B. Bus Topology
 - C. Hybrid Topology
 - D. Full Mesh Topology
40. • What does DTE stand for?
- A. Data Transmission Equipment
 - B. Data Terminal Equipment
 - C. Digital Terminal Exchange
 - D. Data Transfer Environment
41. • Which device directly connects a computer to a network?
- A. Router
 - B. Firewall
 - C. NIC
 - D. Modem
42. • Which network device is closest to end-users and switches data frames?
- A. Router
 - B. Firewall
 - C. Switch
 - D. Gateway
43. • What is a broadcast domain?
- A. A network with no routers
 - B. A set of nodes that can receive broadcast packets
 - C. A wireless communication system
 - D. A fiber optic network
44. • Which device forwards data packets on the Internet?
- A. Switch
 - B. Firewall
 - C. Router
 - D. NIC
45. • What is the main function of a modem?
- A. Encrypt data
 - B. Connect a network to the Internet

- C. Amplify signals
 - D. Store data packets
46. • Which device performs protocol conversion and route selection between networks?
- A. Switch
 - B. Gateway
 - C. Access Point
 - D. NIC
47. • Which network security device monitors and restricts data flow between networks?
- A. Modem
 - B. Router
 - C. Firewall
 - D. Hub
48. • Wi-Fi technology is mainly based on which IEEE standard family?
- A. IEEE 802.3
 - B. IEEE 802.11
 - C. IEEE 802.15
 - D. IEEE 802.16
49. • Which type of access point is commonly used in homes?
- A. Fit AP
 - B. Access Controller
 - C. Fat AP
 - D. Bridge AP
50. • Which wireless device is generally deployed at the aggregation layer of a network?
- A. NIC
 - B. Access Controller
 - C. Modem
 - D. Gateway
51. • Which type of network usually covers only a few square kilometers?
- A. WAN
 - B. MAN
 - C. LAN
 - D. PAN
52. • Which network type connects hosts and LANs within the same city?
- A. LAN
 - B. MAN
 - C. WAN
 - D. SAN
53. • Which network type can span across cities or countries?
- A. LAN
 - B. PAN
 - C. MAN
 - D. WAN
54. • Which OSI layer is closest to the user?
- A. Session Layer
 - B. Application Layer

- C. Network Layer
 - D. Physical Layer
55. • Which OSI layer provides encryption and compression?
- A. Session Layer
 - B. Network Layer
 - C. Presentation Layer
 - D. Data Link Layer
56. • Which OSI layer manages conversations or sessions between applications?
- A. Transport Layer
 - B. Session Layer
 - C. Network Layer
 - D. Physical Layer
57. • Which OSI layer is responsible for end-to-end reliable delivery of data?
- A. Transport Layer
 - B. Application Layer
 - C. Data Link Layer
 - D. Physical Layer
58. • Routers primarily operate at which OSI layer?
- A. Layer 1
 - B. Layer 2
 - C. Layer 3
 - D. Layer 4
59. • Which OSI layer uses MAC addresses for reliable data transfer?
- A. Application Layer
 - B. Data Link Layer
 - C. Session Layer
 - D. Presentation Layer
60. • Which TCP/IP layer is responsible for reliable end-to-end delivery?
- A. Internet Layer
 - B. Network Access Layer
 - C. Application Layer
 - D. Transport Layer
61. • Which topology uses a central hub to connect all devices?
- A. Ring Topology
 - B. Bus Topology
 - C. Star Topology
 - D. Tree Topology
62. • In a bus topology, all devices are connected using a:
- A. Central switch
 - B. Single cable
 - C. Fiber optic ring
 - D. Wireless signal
63. • What is the major disadvantage of a ring topology?
- A. High cable usage
 - B. Difficult to add new nodes

- C. Poor communication speed
 - D. Requires no maintenance
64. • Which topology is also known as hierarchical topology?
- A. Mesh Topology
 - B. Tree Topology
 - C. Bus Topology
 - D. Star Topology
65. • Which topology connects every node to every other node?
- A. Ring Topology
 - B. Star Topology
 - C. Mesh Topology
 - D. Hybrid Topology
66. • Which type of mesh topology connects only some devices to multiple devices?
- A. Full Mesh
 - B. Linear Mesh
 - C. Partial Mesh
 - D. Ring Mesh
67. • What is one advantage of a full mesh topology?
- A. Low cost
 - B. Easy expansion
 - C. High reliability
 - D. Uses less cable
68. • Which topology is formed by combining different topologies?
- A. Bus Topology
 - B. Hybrid Topology
 - C. Ring Topology
 - D. Tree Topology
69. • Which device is considered the end-device in data communication systems?
- A. Router
 - B. Firewall
 - C. Terminal device
 - D. Gateway
70. • Which of the following is an example of DTE?
- A. Router
 - B. Printer
 - C. Firewall
 - D. Access Point
71. • What does NIC stand for?
- A. Network Internet Connector
 - B. Network Interface Card
 - C. Network Internal Cable
 - D. Network Information Converter
72. • Which device is mainly used to switch data frames?
- A. Modem
 - B. Router

- C. Switch
 - D. Firewall
73. • Switches at the access layer are commonly known as:
- A. Layer 1 switches
 - B. Ethernet switches
 - C. Core routers
 - D. Firewalls
74. • What is the primary function of a router?
- A. Store files
 - B. Forward data packets
 - C. Encrypt messages
 - D. Amplify signals
75. • Which device translates ISP signals for local devices?
- A. NIC
 - B. Modem
 - C. Switch
 - D. Firewall
76. • Which device enforces security policies between two networks?
- A. Gateway
 - B. Switch
 - C. Firewall
 - D. Access Point
77. • Which wireless technology is based on IEEE 802.11 standards?
- A. Bluetooth
 - B. WiMAX
 - C. Wi-Fi
 - D. Ethernet
78. • Which access point type is commonly used in homes?
- A. Fit AP
 - B. Fat AP
 - C. Thin AP
 - D. Core AP
79. • Which device manages and configures Fit APs?
- A. Router
 - B. Modem
 - C. Access Controller
 - D. Firewall
80. • Which network type usually connects devices within a campus or building?
- A. WAN
 - B. MAN
 - C. LAN
 - D. PAN
81. • Which network type is designed to connect locations within the same city?
- A. LAN
 - B. MAN

- C. WAN
 - D. SAN
82. • Which network type can span thousands of kilometers?
- A. LAN
 - B. MAN
 - C. WAN
 - D. PAN
83. • Which OSI layer is responsible for providing services to user applications?
- A. Session Layer
 - B. Application Layer
 - C. Transport Layer
 - D. Physical Layer
84. • Which OSI layer handles encryption and compression?
- A. Presentation Layer
 - B. Session Layer
 - C. Network Layer
 - D. Data Link Layer
85. • Which OSI layer controls and ends communication sessions?
- A. Physical Layer
 - B. Transport Layer
 - C. Session Layer
 - D. Application Layer
86. • Which OSI layer ensures reliable end-to-end connectivity?
- A. Data Link Layer
 - B. Network Layer
 - C. Transport Layer
 - D. Physical Layer
87. • Which OSI layer is responsible for routing packets?
- A. Network Layer
 - B. Session Layer
 - C. Presentation Layer
 - D. Physical Layer
88. • Which OSI layer uses MAC addresses?
- A. Network Layer
 - B. Data Link Layer
 - C. Application Layer
 - D. Transport Layer
89. • Which OSI layer defines electrical and mechanical specifications?
- A. Physical Layer
 - B. Session Layer
 - C. Transport Layer
 - D. Presentation Layer
90. • Which TCP/IP layer is also called the host-to-network layer?
- A. Internet Layer
 - B. Transport Layer

- C. Application Layer
- D. Network Access Layer

91. • What does IPv4 stand for?
 - A. Internet Process version 4
 - B. Internet Protocol version 4
 - C. Internal Protocol version 4
 - D. Internet Program version 4
92. • What is the address size of IPv4?
 - A. 16 bits
 - B. 32 bits
 - C. 64 bits
 - D. 128 bits
93. • Which notation is used in IPv4 addresses?
 - A. Hexadecimal notation
 - B. Binary notation
 - C. Dotted decimal notation
 - D. Octal notation
94. • What is the address size of IPv6?
 - A. 32 bits
 - B. 64 bits
 - C. 128 bits
 - D. 256 bits
95. • Which notation is commonly used for IPv6 addresses?
 - A. Decimal notation
 - B. Hexadecimal notation
 - C. Binary notation
 - D. Octal notation
96. • What does MAC stand for in networking?
 - A. Media Access Control
 - B. Machine Access Code
 - C. Main Access Channel
 - D. Media Allocation Control
97. • Which part of the MAC address identifies the manufacturer?
 - A. Host ID
 - B. Vendor ID
 - C. Organizational Unique Identifier (OUI)
 - D. Network ID
98. • How many bits are vendor-specific in the OUI portion of a MAC address?
 - A. 8 bits
 - B. 16 bits
 - C. 24 bits
 - D. 32 bits
99. • Which type of IP address is assigned by an Internet Service Provider?
 - A. Private IP address
 - B. Public IP address

- C. Loopback address
 - D. Multicast address
100. • What type of public IP address changes whenever a device reconnects to the Internet?
- A. Static IP address
 - B. Reserved IP address
 - C. Dynamic IP address
 - D. Loopback IP address
101. • Which part of an IP address identifies the individual device within a network?
- A. Network ID
 - B. Host ID
 - C. Vendor ID
 - D. Gateway ID
102. • Which IPv4 class uses the first octet for the network portion?
- A. Class A
 - B. Class B
 - C. Class C
 - D. Class D
103. • What is the default subnet mask for a Class A IP address?
- A. 255.255.255.0
 - B. 255.255.0.0
 - C. 255.0.0.0
 - D. 255.255.255.255
104. • Which IPv4 class supports multicasting?
- A. Class A
 - B. Class B
 - C. Class D
 - D. Class E
105. • Which IPv4 class is reserved for research purposes?
- A. Class C
 - B. Class D
 - C. Class E
 - D. Class A
106. • Which address range is used for loopback or localhost testing?
- A. 10.0.0.0 – 10.255.255.255
 - B. 127.0.0.0 – 127.255.255.255
 - C. 192.168.0.0 – 192.168.255.255
 - D. 224.0.0.0 – 239.255.255.255
107. • What is the default network address in IPv4?
- A. 255.255.255.255
 - B. 127.0.0.1
 - C. 0.0.0.0
 - D. 169.254.0.1
108. • Which IPv4 address is used for broadcasting messages to the entire network?
- A. 255.255.255.255
 - B. 127.0.0.1

- C. 0.0.0.0
 - D. 192.168.1.1
109. • Which IP address range is commonly used for private home networks?
- A. 224.0.0.0 – 239.255.255.255
 - B. 169.254.0.0 – 169.254.255.255
 - C. 192.168.0.0 – 192.168.255.255
 - D. 127.0.0.0 – 127.255.255.255
110. • What happens if DHCP is unavailable on a network?
- A. Devices shut down automatically
 - B. Devices use link-local addresses
 - C. Devices use multicast addresses
 - D. Devices use Class D addresses
111. • What is the shorthand notation for the default Class C subnet mask?
- A. /8
 - B. /16
 - C. /24
 - D. /32
112. • What is the decimal equivalent of the binary number 00000010?
- A. 1
 - B. 2
 - C. 4
 - D. 8
113. • What logical operation is used in subnet masking?
- A. OR
 - B. XOR
 - C. NOT
 - D. AND
114. • What is subnetting?
- A. Combining networks into one
 - B. Borrowing bits from the host field
 - C. Assigning MAC addresses
 - D. Encrypting IP packets
115. • In subnetting, what is borrowed from the host ID field?
- A. Network ports
 - B. IP addresses
 - C. Bits for subnet IDs
 - D. MAC addresses
116. • Which organization announced IPv4 exhaustion in 2011?
- A. ISO
 - B. IEEE
 - C. IANA
 - D. ICANN
117. • What is another name for IPv6?
- A. IP Classic
 - B. IP Next Generation (IPng)

C. Internet Link Protocol

D. Dynamic IP Protocol

118. • In IPv6 notation, what symbol is used to separate segments?

A. Period (.)

B. Dash (-)

C. Colon (:)

D. Slash (/)

119. • Which IPv6 address is used for loopback testing?

A. ::/128

B. ::1/128

C. FF00::/8

D. 2001::/64

120. • Which IPv6 address type sends packets to all interfaces in a multicast group?

A. Unicast

B. Broadcast

C. Multicast

D. Anycast